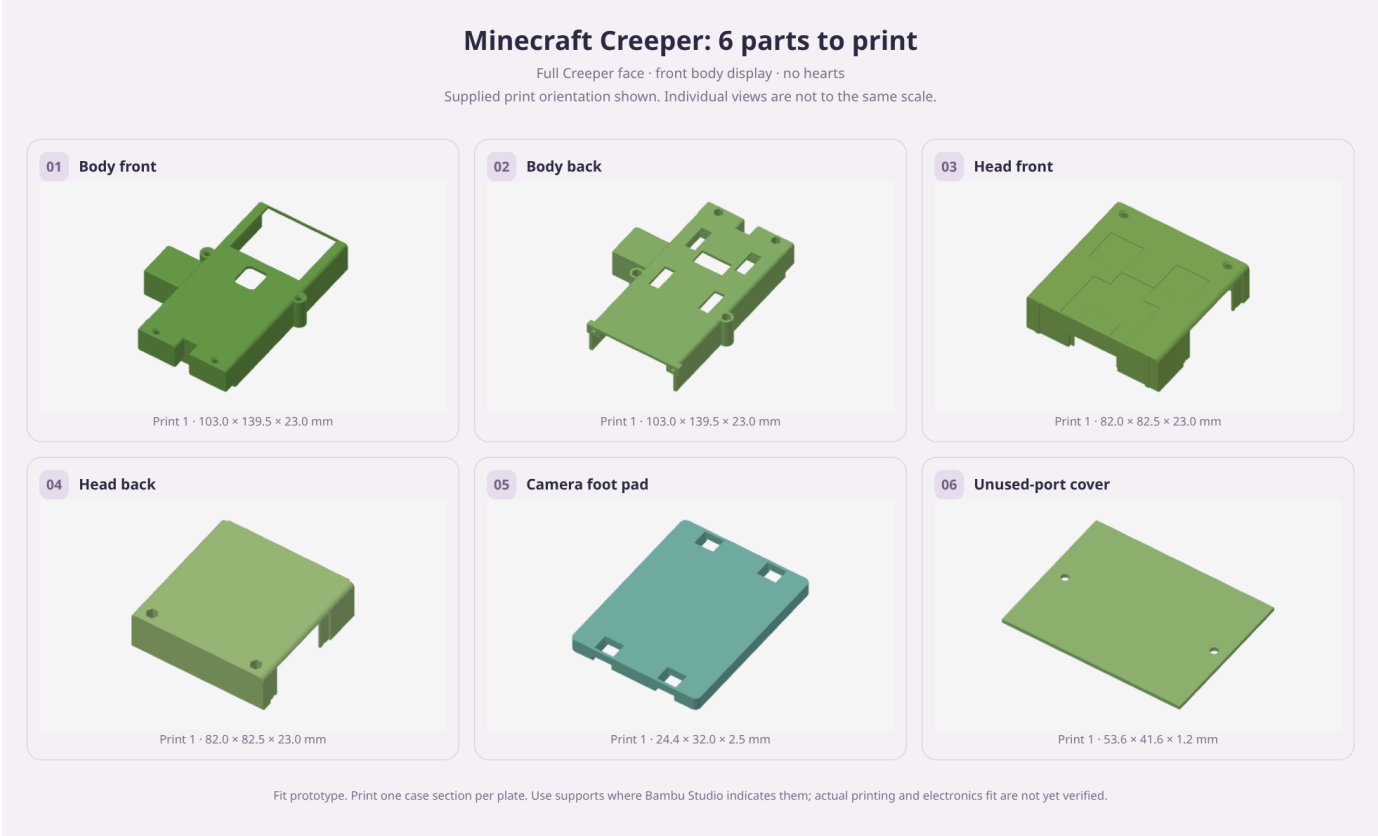
# Minecraft Creeper: parts & assembly

FIT PROTOTYPE · FRONT RED LED DISPLAY · REVERSIBLE SIDE CAMERA · NO HEARTS



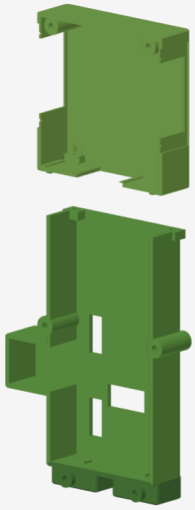

| Hardware (not 3D printed) | Quantity |
|---|---|
| M3 × 30 mm flat-head machine screws | 6 |
| M3 × 16 mm machine screws | 2 |
| Standard M3 hex nuts (5.5 mm across flats) | 8 |
| M2 × 4 mm self-tapping screws for the port cover | 2 |
| Cable ties ≤3.5 mm wide: two ~250 mm + two ~150 mm | 4 |

Print the small hub-footprint test ring first. It checks only the published 59.8 × 82.4 mm footprint, not thickness or connectors. All 6 parts fit the P1S build envelope, but supports and real hardware fit have not been print-tested.

STLs are supplied in their print orientations. Use millimetres and 100% scale; use the matching P1S profile and inspect supports in Bambu Studio. Remove supports before fitting electronics. The original brick tower/base is replaced by this case.

# 1. Join the open back halves

Parts 02 + 04 · 2 × M3×16 screws + nuts

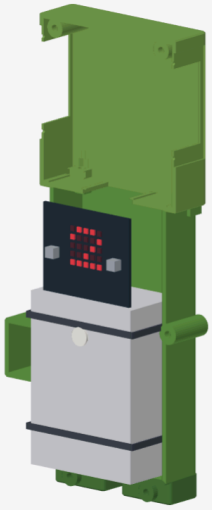


Keep both front covers off. Align the head back above the body back. Join the two small rear neck brackets: the screws go up from the recessed underside, with nuts in the head-side pockets. Joining the backs first leaves access to the internal fasteners.

Power off/unplug before fitting. Check the real hub depth and connector position before strapping it in.

# 2. Fit the hub and micro:bit

Upright hub · two long cable ties

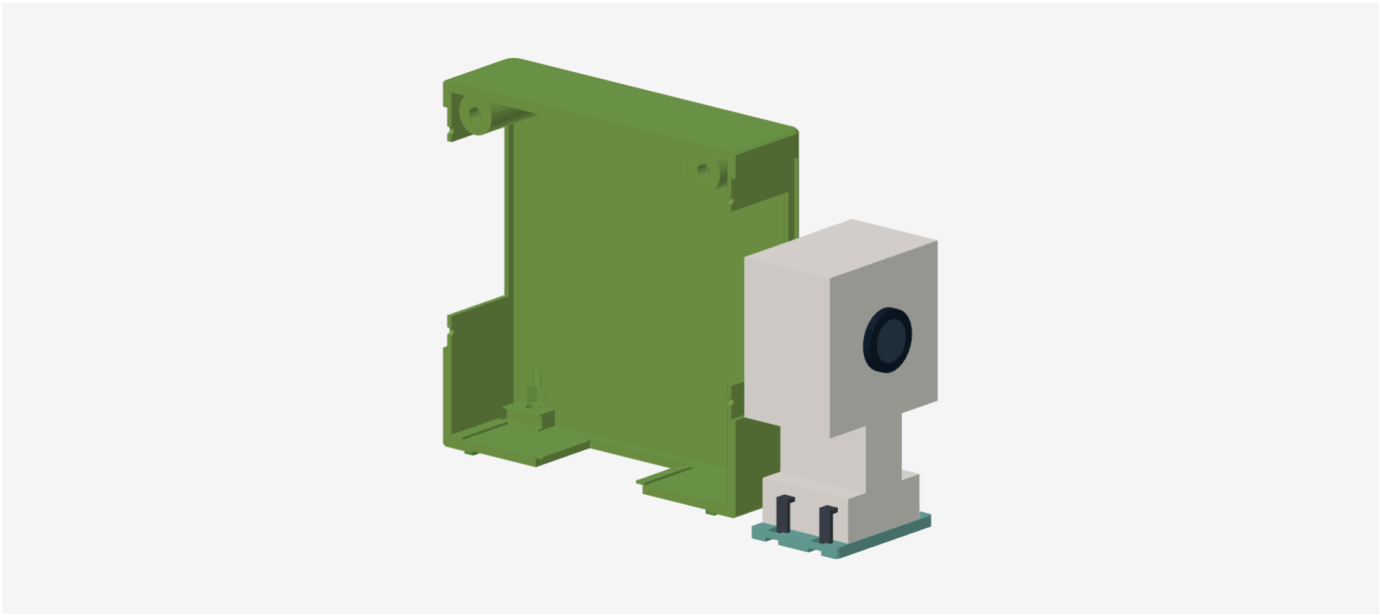


Keep the micro:bit connected to the hub. Rest the unit on the four small floor pads and align the red LED matrix with the body opening. Secure the hub with two long cable ties through the rear slots, choosing positions clear of the real ports and switch. Cut no wires or board parts; stop if the reference layout does not match your hardware.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

# 3. Prepare the camera on either side

Part 05 · two small cable ties

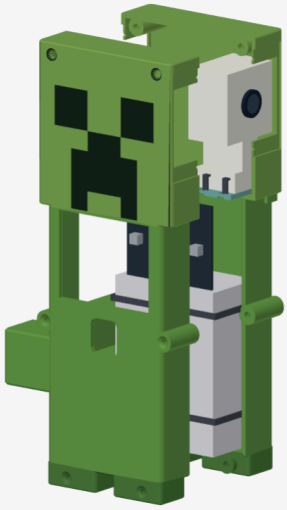


The right side is illustrated. Put the camera mounting foot on the pad and thread two small ties through the pad slots and underside grooves. Seat the pad on the head floor supports. For the left side, rotate the same pad 180° and move it to the other pocket. Keep the lens clear. Connect the existing camera lead to an IIC port on the hub and route it through the neck with slack.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

# 4. Close the front panels

Parts 01 + 03 · 6 × M3×30 flat-head screws + nuts

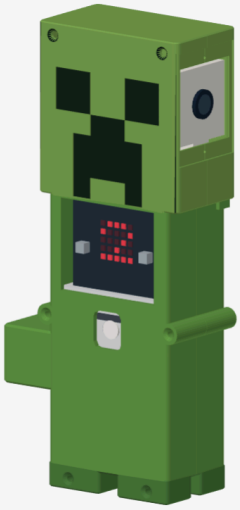


Dry-fit the body front and head front while watching the cable and display. Insert M3 nuts into the rear-entry hex channels. Fit four screws to the body (two in the feet and two in the small side bosses), then two at the top of the head. Do not force a joint that will not seat; the cable, camera pad or electronics may need repositioning.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

# 5. Cover the unused port and test

Part 06 · 2 × M2×4 self-tapping screws

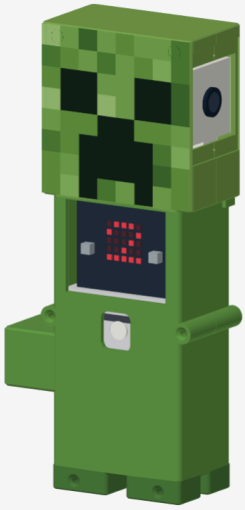


Put the flush cover on the unused camera side and secure it in the small pilot holes with two short M2 self-tapping screws. Test the whole 5×5 display, A/B buttons, camera recognition and power/charging access before final tightening. Swapping the camera side later requires opening the head, moving the camera/pad and moving the cover.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.

# 6. Paint the Minecraft pixels

Use the included pixel-colour reference



Use green filament or paint the case green. Paint the engraved eyes and frown black, then add lighter and darker green pixel patches using the colour reference. Keep paint off the lens, display, buttons, mating faces and screw channels. The STL files contain shape only; the mottled preview colours are a painting or multicolour-printing example.

Do not force a fit or permanently glue the case. This layout still uses an assumed hub depth and camera thickness.